

# WEEK-2

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## Step 1: Create Database

The screenshot shows the SQL Developer interface. On the left, the 'SCHEMAS' pane lists 'bookflow\_db', 'practice', and 'sys'. The main editor window, titled 'WEEK1\* ddl SQL File 4\*', contains the SQL statement: `CREATE DATABASE bookflow_week2_db;`. The 'Output' pane at the bottom shows the execution results: a single row with a green checkmark, indicating success. The message states: '1 14:43:17 CREATE DATABASE bookflow\_week2\_db 1 row(s) affected'.

## Step 2: Create Books table and insert books

The screenshot shows the SQL Developer interface with the 'SCHEMAS' pane on the left. The main editor window contains the following SQL statements: `CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100), isbn VARCHAR(20), published_year INT);` followed by an `INSERT INTO` statement with three rows of book data. Below the editor, the 'Result Grid' displays the inserted data in a table format. The 'Output' pane at the bottom shows the execution results for all steps, including the table creation and the insertion of three rows.

| book_id | title                 | isbn          | published_year |
|---------|-----------------------|---------------|----------------|
| 1       | The Great Gatsby      | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                  | 9780451524935 | 1949           |

The 'Output' pane shows the following execution results:

| # | Time     | Action                                         | Message                                        | Duration / Fets |
|---|----------|------------------------------------------------|------------------------------------------------|-----------------|
| 1 | 14:43:17 | CREATE DATABASE bookflow_week2_db              | 1 row(s) affected                              | 0.000 sec       |
| 2 | 14:44:10 | USE bookflow_week2_db                          | 0 row(s) affected                              | 0.000 sec       |
| 3 | 14:44:13 | CREATE TABLE Books ( book_id INT PRI...        | 0 row(s) affected                              | 0.094 sec       |
| 4 | 14:44:31 | SELECT * FROM Books LIMIT 0, 1000              | 0 row(s) returned                              | 0.000 sec / 0   |
| 5 | 14:44:38 | INSERT INTO Books (book_id, title, isbn, pu... | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.047 sec       |
| 6 | 14:44:48 | SELECT * FROM Books LIMIT 0, 1000              | 3 row(s) returned                              | 0.000 sec / 0   |

### Step 3: Create Table for members and insert members

The screenshot shows the SQL Developer interface. The left pane displays the 'SCHEMAS' tree with 'bookflow\_db', 'practice', and 'sys' schemas. The main editor shows the following SQL script:

```
18 email VARCHAR(100) UNIQUE
19 );
20 • INSERT INTO Members (member_id, full_name, email) VALUES
21 (101, 'Karuturi Sai Akhil', 'saiaakhil@email.com'),
22 (102, 'Virat Kohli', 'viratkohli18@email.com'),
23 (103, 'MS Dhoni', 'msdhoni07@email.com');
24 • SELECT * FROM Members;
```

The 'Result Grid' below the editor displays the data inserted into the 'Members' table:

| member_id | full_name          | email                  |
|-----------|--------------------|------------------------|
| 101       | Karuturi Sai Akhil | saiaakhil@email.com    |
| 102       | Virat Kohli        | viratkohli18@email.com |
| 103       | MS Dhoni           | msdhoni07@email.com    |

The bottom pane shows the 'Output' window with the 'Action Output' tab selected, displaying the execution log:

| # | Time     | Action                                        | Message                                        | Duration / |
|---|----------|-----------------------------------------------|------------------------------------------------|------------|
| 4 | 14:44:31 | SELECT * FROM Books LIMIT 0, 1000             | 0 row(s) returned                              | 0.000 sec  |
| 5 | 14:44:38 | INSERT INTO Books (book_id, title, isbn, p... | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.047 sec  |
| 6 | 14:44:48 | SELECT * FROM Books LIMIT 0, 1000             | 3 row(s) returned                              | 0.000 sec  |
| 7 | 14:45:22 | CREATE TABLE Members ( member_id l...         | 0 row(s) affected                              | 0.015 sec  |
| 8 | 14:46:21 | INSERT INTO Members (member_id, full_n...     | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.016 sec  |
| 9 | 14:46:26 | SELECT * FROM Members LIMIT 0, 1000           | 3 row(s) returned                              | 0.000 sec  |

### Step 4: CREATE LOANS TABLE

The screenshot shows the SQL Developer interface. The left pane displays the 'SCHEMAS' tree. The main editor shows the following SQL script:

```
16 member_id INT PRIMARY KEY,
17 full_name VARCHAR(100),
18 email VARCHAR(100) UNIQUE
19 );
20 • INSERT INTO Members (member_id, full_name, email) VALUES
21 (101, 'Karuturi Sai Akhil', 'saiaakhil@email.com'),
22 (102, 'Virat Kohli', 'viratkohli18@email.com'),
23 (103, 'MS Dhoni', 'msdhoni07@email.com');
24 • SELECT * FROM Members;
25
26 • CREATE TABLE Loans (
27 loan_id INT PRIMARY KEY,
28 member_id INT,
29 book_id INT,
30 loan_date DATE,
31 FOREIGN KEY (member_id) REFERENCES Members(member_id),
32 FOREIGN KEY (book_id) REFERENCES Books(book_id)
33 );
34
```

The bottom pane shows the 'Output' window with the 'Action Output' tab selected, displaying the execution log:

| #  | Time     | Action                                        | Message                                        | Duration / Fetch      |
|----|----------|-----------------------------------------------|------------------------------------------------|-----------------------|
| 5  | 14:44:38 | INSERT INTO Books (book_id, title, isbn, p... | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.047 sec             |
| 6  | 14:44:48 | SELECT * FROM Books LIMIT 0, 1000             | 3 row(s) returned                              | 0.000 sec / 0.000 sec |
| 7  | 14:45:22 | CREATE TABLE Members ( member_id l...         | 0 row(s) affected                              | 0.015 sec             |
| 8  | 14:46:21 | INSERT INTO Members (member_id, full_n...     | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.016 sec             |
| 9  | 14:46:26 | SELECT * FROM Members LIMIT 0, 1000           | 3 row(s) returned                              | 0.000 sec / 0.000 sec |
| 10 | 14:47:01 | CREATE TABLE Loans ( loan_id INT PR...        | 0 row(s) affected                              | 0.031 sec             |

Step 5: INSERT RECORDS INTO LOANS TABLE

Navigator

SCHEMAS

- bookflow\_db
- practice
- sys

WEEK1\* ddl SQL File 4\*

```
39 (5, 102, 1, '2025-02-05'),
40 (6, 103, 2, '2025-02-12'),
41 (7, 101, 3, '2025-03-01'),
42 (8, 102, 3, '2025-03-07'),
43 (9, 103, 1, '2025-03-15'),
44 (10, 101, 1, '2025-04-01');
45 SELECT * FROM Loans;
```

Result Grid

| loan_id | member_id | book_id | loan_date  |
|---------|-----------|---------|------------|
| 1       | 101       | 1       | 2025-01-05 |
| 2       | 102       | 2       | 2025-01-08 |
| 3       | 103       | 3       | 2025-01-10 |
| 4       | 101       | 2       | 2025-02-01 |
| 5       | 102       | 1       | 2025-02-05 |
| 6       | 103       | 2       | 2025-02-12 |
| 7       | 101       | 3       | 2025-03-01 |
| 8       | 102       | 3       | 2025-03-07 |
| 9       | 103       | 1       | 2025-03-15 |
| 10      | 101       | 1       | 2025-04-01 |

Loans 4 x

Output

Action Output

| #  | Time     | Action                                     | Message                                        | Duration / Fetch      |
|----|----------|--------------------------------------------|------------------------------------------------|-----------------------|
| 7  | 14:45:22 | CREATE TABLE Members ( member_id I...      | 0 row(s) affected                              | 0.015 sec             |
| 8  | 14:46:21 | INSERT INTO Members (member_id, full_n...  | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.016 sec             |
| 9  | 14:46:26 | SELECT * FROM Members LIMIT 0, 1000        | 3 row(s) returned                              | 0.000 sec / 0.000 sec |
| 10 | 14:47:01 | CREATE TABLE Loans ( loan_id INT PR...     | 0 row(s) affected                              | 0.031 sec             |
| 11 | 14:47:25 | INSERT INTO Loans (loan_id, member_id, ... | 10 row(s) affected Records: 10 Duplicates:...  | 0.016 sec             |
| 12 | 14:47:32 | SELECT * FROM Loans LIMIT 0, 1000          | 10 row(s) returned                             | 0.000 sec / 0.000 sec |

Step 6: JOIN QUERY

Navigator

SCHEMAS

- bookflow\_db
- practice
- sys

WEEK1\* ddl SQL File 4\*

```
43 (9, 103, 1, '2025-03-15'),
44 (10, 101, 1, '2025-04-01');
45 SELECT * FROM Loans;
46 SELECT m.full_name AS Member_Name, b.title AS Book_Title
47 FROM Loans l
48 INNER JOIN Members m ON l.member_id = m.member_id
49 INNER JOIN Books b ON l.book_id = b.book_id;
```

Result Grid

| Member_Name        | Book_Title            |
|--------------------|-----------------------|
| Karuturi Sai Akhli | The Great Gatsby      |
| Karuturi Sai Akhli | To Kill a Mockingbird |
| Karuturi Sai Akhli | 1984                  |
| Karuturi Sai Akhli | The Great Gatsby      |
| Virat Kohli        | To Kill a Mockingbird |
| Virat Kohli        | The Great Gatsby      |
| Virat Kohli        | 1984                  |
| MS Dhoni           | 1984                  |
| MS Dhoni           | To Kill a Mockingbird |
| MS Dhoni           | The Great Gatsby      |

Result 5 x

Output

Action Output

| #  | Time     | Action                                     | Message                                        | Duration / Fetch      |
|----|----------|--------------------------------------------|------------------------------------------------|-----------------------|
| 8  | 14:46:21 | INSERT INTO Members (member_id, full_n...  | 3 row(s) affected Records: 3 Duplicates: 0 ... | 0.016 sec             |
| 9  | 14:46:26 | SELECT * FROM Members LIMIT 0, 1000        | 3 row(s) returned                              | 0.000 sec / 0.000 sec |
| 10 | 14:47:01 | CREATE TABLE Loans ( loan_id INT PR...     | 0 row(s) affected                              | 0.031 sec             |
| 11 | 14:47:25 | INSERT INTO Loans (loan_id, member_id, ... | 10 row(s) affected Records: 10 Duplicates:...  | 0.016 sec             |
| 12 | 14:47:32 | SELECT * FROM Loans LIMIT 0, 1000          | 10 row(s) returned                             | 0.000 sec / 0.000 sec |
| 13 | 14:48:47 | SELECT m.full_name AS Member_Name, b...    | 10 row(s) returned                             | 0.000 sec / 0.000 sec |

Step 7: GROUP BY + COUNT

**SCHEMAS**

Filter objects

- bookflow\_db
- practice
- sys

**SQL Editor**

```

48 INNER JOIN Members m ON l.member_id = m.member_id
49 INNER JOIN Books b ON l.book_id = b.book_id;
50
51 • SELECT published_year, COUNT(book_id) AS Total_Books
52 FROM Books
53 GROUP BY published_year
54 ORDER BY published_year;

```

**Result Grid**

| published_year | Total_Books |
|----------------|-------------|
| 1925           | 1           |
| 1949           | 1           |
| 1960           | 1           |

**Administration Schemas**

Information

No object selected

**Object Info Session**

**Result 6**

Output

Action Output

| #  | Time     | Action                                     | Message                                      | Dura  |
|----|----------|--------------------------------------------|----------------------------------------------|-------|
| 9  | 14:46:26 | SELECT * FROM Members LIMIT 0, 1000        | 3 row(s) returned                            | 0.000 |
| 10 | 14:47:01 | CREATE TABLE Loans ( loan_id INT PR...     | 0 row(s) affected                            | 0.031 |
| 11 | 14:47:25 | INSERT INTO Loans (loan_id, member_id, ... | 10 row(s) affected Records: 10 Duplicates... | 0.016 |
| 12 | 14:47:32 | SELECT * FROM Loans LIMIT 0, 1000          | 10 row(s) returned                           | 0.000 |
| 13 | 14:48:47 | SELECT m.full_name AS Member_Name, b...    | 10 row(s) returned                           | 0.000 |
| 14 | 14:49:32 | SELECT published_year, COUNT(book_id) ...  | 3 row(s) returned                            | 0.016 |

## Step 8: Display Donation History

**SCHEMAS**

Filter objects

- bookflow\_db
- practice
- sys

**SQL Editor**

```

67 VALUES (4, 'Animal Farm', '9780451526342', 1945);
68 • INSERT INTO Donation_History
69 (donation_id, book_id, donor_name, donation_date)
70 VALUES
71 (1, 4, 'Karuturi Sai Akhil', CURDATE());
72 • COMMIT;
73 • SELECT * FROM Donation_History;

```

**Result Grid**

| donation_id | book_id | donor_name         | donation_date |
|-------------|---------|--------------------|---------------|
| 1           | 4       | Karuturi Sai Akhil | 2026-08-10    |

**Administration Schemas**

Information

No object selected

**Donation\_History 12**

Output

Action Output

| #  | Time     | Action                                        | Message           | Dura  |
|----|----------|-----------------------------------------------|-------------------|-------|
| 24 | 14:54:20 | START TRANSACTION                             | 0 row(s) affected | 0.000 |
| 25 | 14:54:28 | INSERT INTO Books (book_id, title, isbn, p... | 1 row(s) affected | 0.000 |
| 26 | 14:54:45 | INSERT INTO Donation_History (donation_...    | 1 row(s) affected | 0.000 |
| 27 | 14:54:51 | COMMIT                                        | 0 row(s) affected | 0.000 |
| 28 | 14:54:56 | SELECT * FROM Books LIMIT 0, 1000             | 4 row(s) returned | 0.000 |
| 29 | 14:55:01 | SELECT * FROM Donation_History LIMIT 0...     | 1 row(s) returned | 0.000 |

## Step 9: Create ISBN Index + Search

Navigator

SCHEMAS

Filter objects

bookflow\_db

practice

sys

Administration

Schemas

Information

No object selected

Object InfoSession

WEEK1\*ddlSQL File 4\*

Limit to 1000 rows

```
72 • COMMIT;
73 • SELECT * FROM Donation_History;
74
75 • CREATE INDEX idx_books_isbn ON Books(isbn);
76 • SELECT *
77 FROM Books
78 WHERE isbn = '9780451524935';
```

Result Grid

Filter Rows:

EditExport/Import

|   | book_id | title | isbn          | published_year |
|---|---------|-------|---------------|----------------|
| ▶ | 3       | 1984  | 9780451524935 | 1949           |
| * | NULL    | NULL  | NULL          | NULL           |

Books 13

Output

Action Output

| #    | Time     | Action                                     | Message                                |
|------|----------|--------------------------------------------|----------------------------------------|
| ✓ 26 | 14:54:45 | INSERT INTO Donation_History (donation_... | 1 row(s) affected                      |
| ✓ 27 | 14:54:51 | COMMIT                                     | 0 row(s) affected                      |
| ✓ 28 | 14:54:56 | SELECT * FROM Books LIMIT 0, 1000          | 4 row(s) returned                      |
| ✓ 29 | 14:55:01 | SELECT * FROM Donation_History LIMIT 0...  | 1 row(s) returned                      |
| ✓ 30 | 14:55:55 | CREATE INDEX idx_books_isbn ON Book...     | 0 row(s) affected Records: 0 Duplicate |
| ✓ 31 | 14:56:02 | SELECT * FROM Books WHERE isbn = '...      | 1 row(s) returned                      |